KG 571: Foundations of Programming
Assignment 3: Logical Analysis

Submit your work in a single pdf/doc format and submit.

Part A: Applied Logical Analysis

Task 1: Truth Tables & Classification

For each statement below:

1. Build a complete truth table
2. Classify it as a **tautology**, **contradiction**, or **contingency**

- $(p \vee q) \rightarrow r$
- $(p \wedge q) \vee (p \wedge r)$
- $(p \vee q) \equiv (p \wedge q)$

Task 2: Logical Equivalence

Use **truth tables or logical laws** to determine if the following pairs are equivalent. Show brief work.

- $(p \vee q)$ and $(p \rightarrow q)$
- $(p \wedge q)$ and $(p \vee q)$
- $(p \wedge \neg q) \vee (\neg p \wedge q)\ and\ (p \vee q) \wedge \neg(p \wedge q)$

Part B: Short Answers – Pretest & Posttest

Answer briefly (3–6 sentences each).

- What is a **pretest loop**? Provide one small pseudocode example and explain when it is useful.
- What is a **posttest loop**? Provide one example and explain when it is preferred.
- Compare pretest vs. posttest loops. Which one guarantees at least one execution?
- Give one real-world scenario where a posttest loop works better than a pretest loop. Explain why.